# MT8002 Elements of Set Theory
# Chapter 7. Ordered Sets
# Section 7.4. Well-Ordered Sets

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Semester 2026-2

# Preliminaries

## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

Well-Ordered Sets
Some Properties of Well-Ordered Sets
Comparing Well-Ordered Sets
Initial Segments and Proof by Induction
References

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Well-Ordered Sets
Some Properties of Well-Ordered Sets
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References

# Well-Ordered Sets

## Definition (p. 188)

Let $(X, <)$ be a linearly ordered set. The relation $<$ is a **well-order** (or the set $X$ is **well-ordered** by the relation $<$) if every non-empty subset $A$ of $X$ has a least element, i.e., there is some element $a \in A$ such that $a \leq x$, for all $x \in A$.

# Well-Ordered Sets

## Definition (p. 188)

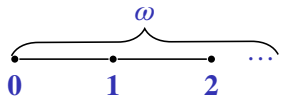
Let $(X, <)$ be a linearly ordered set. The relation $<$ is a **well-order** (or the set $X$ is **well-ordered** by the relation $<$) if every non-empty subset $A$ of $X$ has a least element, i.e., there is some element $a \in A$ such that $a \leq x$, for all $x \in A$.[†]

[†]Note that the terms 'well-order' relation and 'well-ordered' set are not standard in the literature. For example, Enderton (1977) uses 'well ordering' for relations and 'well ordered' for sets and Button (2021) uses 'well-ordering' for both relations and sets.

# Well-Ordered Sets

## Example

The canonical example of a well-ordered set is $\omega$, i.e., $(\mathbb{N}, \in)$,




because every non-empty subset of $\omega$ has a least element (Theorem 3.9).

# Well-Ordered Sets

## Example (informal)

The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

# Well-Ordered Sets

## Example (informal)

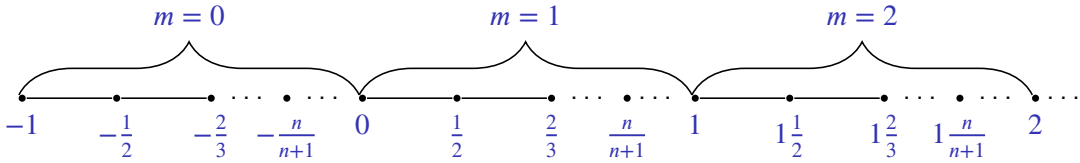
The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

Let $a, b \in \mathbb{R}$, with $a < b$. In general, any subset of $\mathbb{R}$ containing an open interval $(a, b)$ is not well-ordered.

# Well-Ordered Sets

## Example (informal)

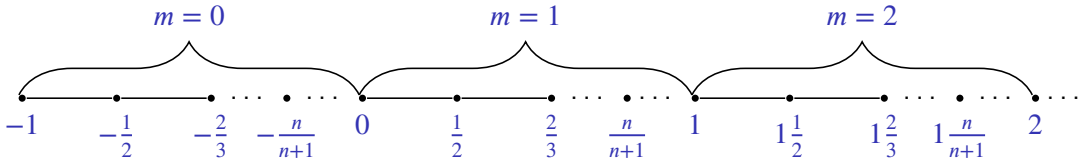
The subset $X = \left\{ m - \frac{1}{n+1} : m, n \in \mathbb{N} \right\}$ of $\mathbb{Q}$ with the usual order is well-ordered.



# Well-Ordered Sets

## Example (informal)

The subset $X = \left\{ m - \frac{1}{n+1} : m, n \in \mathbb{N} \right\}$ of $\mathbb{Q}$ with the usual order is well-ordered.




Note that

$$m - \frac{1}{n+1} < m' - \frac{1}{n'+1} \text{ iff } \begin{cases} m < m', \text{ or} \\ m = m' \text{ and } n < n'. \end{cases}$$

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## Example (continuation)

Let $A$ a non-empty subset of $X$.

"*The least element of $A$ has to have the least $m$ available among the elements of $A$ and the smallest $n$ for this $m$.*" *(p. 189)*

We find the least element of $A$ using the well-order property of natural numbers.

Let $B = \left\{ m \in \mathbb{N} : m - \frac{1}{n+1} \in A, \text{ for some } n \in \mathbb{N} \right\}$ and let $m_0$ the least element of $B$.

Let $C = \left\{ n \in \mathbb{N} : m_0 - \frac{1}{n+1} \in A \right\}$ and let $n_0$ the least element of $C$.

Then $m_0 - \frac{1}{n_0+1}$ is the least element of $A$.

# Outline

Well-Ordered Sets
Some Properties of Well-Ordered Sets
Comparing Well-Ordered Sets
Initial Segments and Proof by Induction
References

# Some Properties of Well-Ordered Sets

## Theorem

Let $(X, <)$ be a well-ordered set and let $A \subseteq X$. Then the set $A$ is well-ordered by the restriction of relation $<$ to $A$, i.e., $(A, <|_{A \times A})$ is a well-ordered set (Exercise 7.44).

# Some Properties of Well-Ordered Sets

## Example

The set $\mathbf{5} \doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}$ is well-ordered by the relation $\in$.

# Some Properties of Well-Ordered Sets

Let $X$ be a set. Recall that an infinite descending $\in$-chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

# Some Properties of Well-Ordered Sets

Let $X$ be a set. Recall that an infinite descending $\in$-chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

## Definition (p. 190)

Let $(X, <)$ be an ordered set. An **infinite descending $<$-chain** is a sequence $\langle x_n \rangle$ such that $x_{n+1} < x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} < x_n < \cdots < x_2 < x_1 < x_0.$$

# Some Properties of Well-Ordered Sets

## Theorem

Let $(X, <)$ be a well-ordered set and suppose that $X$ contains an infinite descending $<$-chain. Then $X$ is not well-ordered by $<$ (Exercise 7.45).

# Some Properties of Well-Ordered Sets

## Example (informal)

The set $\mathbb{Z}$ with the usual order $<$ is not a well-ordered set because the subset of negative integers forms an infinite descending $<$-chain.

$$\cdots < -3 < -2 < -1$$

# Some Properties of Well-Ordered Sets

## Theorem

Let $X$ and $Y$ be ordered sets and let $f : X \longrightarrow Y$ be an order-isomorphism. Then if $X$ is well-ordered then so is $Y$ (Exercise 7.46).

# Some Properties of Well-Ordered Sets

## Example

The set $\omega^2$ is well-ordered because there is an order-isomorphism between it and the subset $\left\{m - \frac{1}{n+1} : m, n \in \mathbb{N}\right\}$ of $\mathbb{Q}$ with the usual order.

# Some Properties of Well-Ordered Sets

## Exercise 7.47

Suppose that $X$ and $Y$ are well-ordered sets. Show that the ordered sets $X + Y$ and $X \times Y$ are both well-ordered.

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# Comparing Well-Ordered Sets

## Comparison of ordered sets

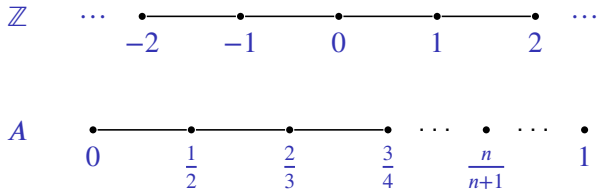
Two ordered sets are **comparable** if either they are order-isomorphic or one can be order-embedded into the other.

Any two well-ordered sets are comparable, but, two linearly ordered sets are not necessarily comparable.

# Comparing Well-Ordered Sets

## Example (informal)

The sets $\mathbb{Z}$ and $A = \left\{1 - \frac{1}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}$, both with the usual order $<$, are incomparable (Exercise 7.50).




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# Comparing Well-Ordered Sets

## Example (continuation)

(a) $\mathbb{Z} \ncong A$ because $1$ is a point limit of $A$ but $\mathbb{Z}$ has not limit points.

(b) There is not an order-embedding $f : \mathbb{Z} \longrightarrow A$ because $\mathbb{Z}$ contains an infinite descending $<$-chain but $A$ is a well-ordered set, i.e., $A$ does not contain an infinite descending $<$-chain. Therefore we cannot order-embedding $\mathbb{Z}$ into $A$.

(c) There is not an order-embedding $f : A \longrightarrow \mathbb{Z}$ because "*whichever integers we chose for $f(0)$ and $f(1)$, there would only be finitely many possible images between them for the infinitely many $1 - \frac{1}{n+1}$, $n > 0$. So we could not define $f$ to be both one-one and order-preserving.*" *(p. 191)*

# Comparing Well-Ordered Sets

## Comparison of well-ordered sets

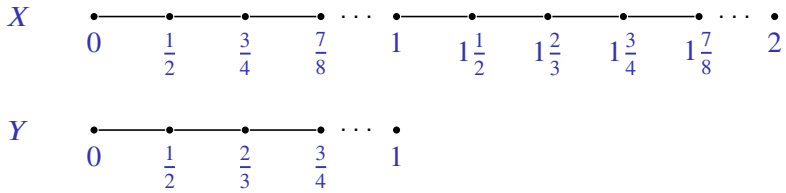
"*The basic idea is to pair off the least elements of $X$ and $Y$, then their next least elements, and so on. If the process has worked for all the elements of $X$ less than some limit point $x$ of $X$, it turns out that the least available element of $Y$ to be paired up with $x$ also has to be a limit point. And if in this process we run out of elements of both $X$ and $Y$ at the same stage, we have an order-isomorphism; while if we run out of elements of one of the sets before we've used up all the elements of the other, then we have an order-embedding of one set into the other.*" *(p. 192)*

# Comparing Well-Ordered Sets

## Example (informal)

Let $X$ and $Y$ be well-ordered sets, subsets of $\mathbb{Q}$, with the usual order $<$, defined by

$$X = \left\{1 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \left\{2 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \{2\},$$
$$Y = \left\{1 - \frac{1}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}.$$



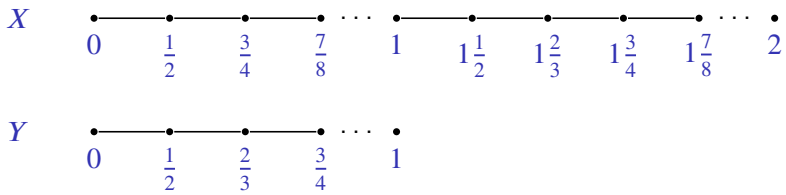

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# Comparing Well-Ordered Sets

## Example (informal)

Let $X$ and $Y$ be well-ordered sets, subsets of $\mathbb{Q}$, with the usual order $<$, defined by[†]

$$X = \left\{1 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \left\{2 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \{2\},$$
$$Y = \left\{1 - \frac{1}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}.$$

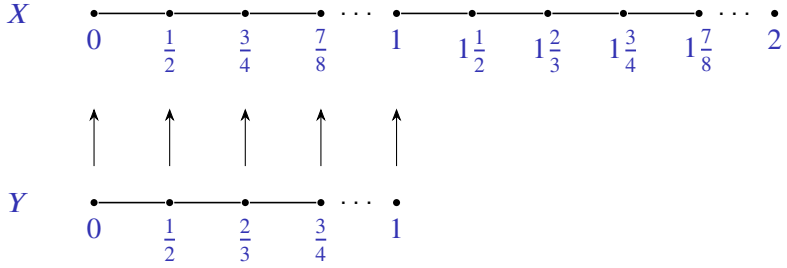



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[†]See the corrections to the textbook in the course homepage.

# Comparing Well-Ordered Sets

## Example (continuation)




The set $Y$ runs out of elements before the set $X$, giving an order-embedding

$$\begin{aligned} f : Y &\longrightarrow X \\ 1 - \frac{1}{n+1} &\longmapsto 1 - \frac{1}{2^n}, \quad n \in \mathbb{N}, \\ 1 &\longmapsto 1. \end{aligned}$$

# Comparing Well-Ordered Sets

## Definition (p. 193)

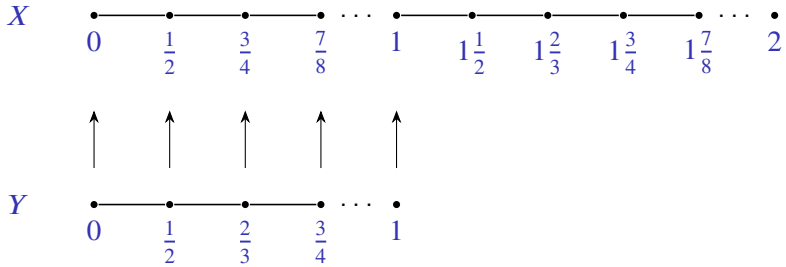
Let $(X, <)$ be a well-ordered and let $a \in X$. The **initial segment of $X$ determined by $a$**, denoted $\mathrm{Seg}_X(a)$, is the set defined by

$$\mathrm{Seg}_X(a) := \{x \in X : x < a\}.$$

# Comparing Well-Ordered Sets

## Example (informal)

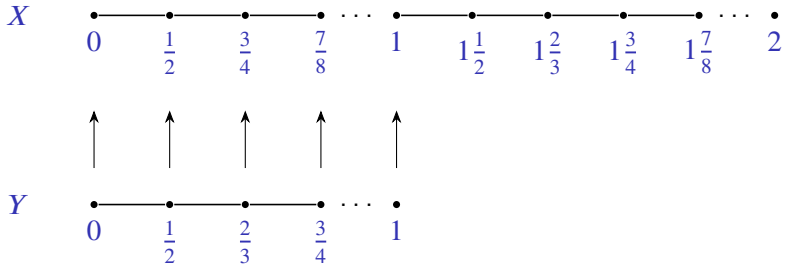
The order-embedding from the previous example maps the set $Y$ onto the initial segment $\mathrm{Seg}_X(1\frac{1}{2})$ of the set $X$.



# Comparing Well-Ordered Sets

## Example (informal)

The order-embedding from the previous example maps the set $Y$ onto the initial segment $\mathrm{Seg}_X(1\frac{1}{2})$ of the set $X$.†




†See the corrections to the textbook in the course homepage.

# Comparing Well-Ordered Sets

## Example

The initial segment of $\omega$ determined by $\mathbf{3}$ is the set

$$\begin{aligned}\mathrm{Seg}_{\omega}(\mathbf{3}) &\doteq \{\mathbf{n} \in \omega : \mathbf{n} \in \mathbf{3}\} \\ &\doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}\} \\ &\doteq \mathbf{3}.\end{aligned}$$

# Comparing Well-Ordered Sets

## Theorem 7.5

Let $X$ and $Y$ be well-ordered sets. Then exactly one of the following holds:

(a) the set $X$ is order-isomorphic to an initial segment of the set $Y$,

(b) the set $X$ is order-isomorphic to the set $Y$,

(c) the set $Y$ is order-isomorphic to an initial segment of the set $X$.

# Comparing Well-Ordered Sets

## Theorem 7.5

Let $X$ and $Y$ be well-ordered sets. Then exactly one of the following holds:

(a) the set $X$ is order-isomorphic to an initial segment of the set $Y$,

(b) the set $X$ is order-isomorphic to the set $Y$,

(c) the set $Y$ is order-isomorphic to an initial segment of the set $X$.

"*Theorem 7.5 is a very strong result: any two well-orders are essentially the same [i.e., order-isomorphic], or one is just the other with some extra elements tacked on at the end. It also suggests that one can organise well-ordered sets in terms of their 'length' as orders.*" *(p. 194)*

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# Initial Segments and Proof by Induction

**Theorem 3.5 (Proof by induction on $\mathbb{N}$)**

For all subsets $A \subseteq \mathbb{N}$ such that:

(a) Base step: $\mathbf{0} \in A$;

(b) Inductive step: if $\mathbf{n} \in A$ then $\mathbf{n}^+ \in A$.

Then $A \doteq \mathbb{N}$.

# Initial Segments and Proof by Induction

## Theorem 3.5 (Proof by induction on $\mathbb{N}$)

For all subsets $A \subseteq \mathbb{N}$ such that:

(a) Base step: $\mathbf{0} \in A$;

(b) Inductive step: if $\mathbf{n} \in A$ then $\mathbf{n}^+ \in A$.

Then $A \doteq \mathbb{N}$.

## Theorem 7.6 (Principle of transfinite induction)

Let $X$ be a well-ordered set. For all subsets $A \subseteq X$ such that:

(a) Base step: the least element of $X$ is in $A$;

(b) Inductive step: for all other $x \in X$, if $\mathrm{Seg}_X(x) \subseteq A$ then $x \in A$.

Then $A \doteq X$.

## Technical remark

Note that the textbook excludes the least element of the set $X$ from the inductive step of Theorem 7.6 because this element triviality satisfies the condition.

# Initial Segments and Proof by Induction

## Theorem 7.7

Let $(X, <)$ be a well-ordered set and let $f : X \longrightarrow X$ be an order-embedding of $X$ into itself. Then $x \leq f(x)$, for all $x \in X$.

# Initial Segments and Proof by Induction

## Theorem 7.8

Let $(X, <)$ be a well-ordered set and let $f : X \longrightarrow X$ be an order-embedding of $X$ into itself. Then $f$ is the identity function.

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# References

Tim Button (2021). Set Theory. An Open Introduction. Fall 2021. Revision 2138a50 (2026-07-11). Independently published. URL: https://builds.openlogicproject.org/courses/set-theory/ (cit. on p. 6).

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 6).

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).